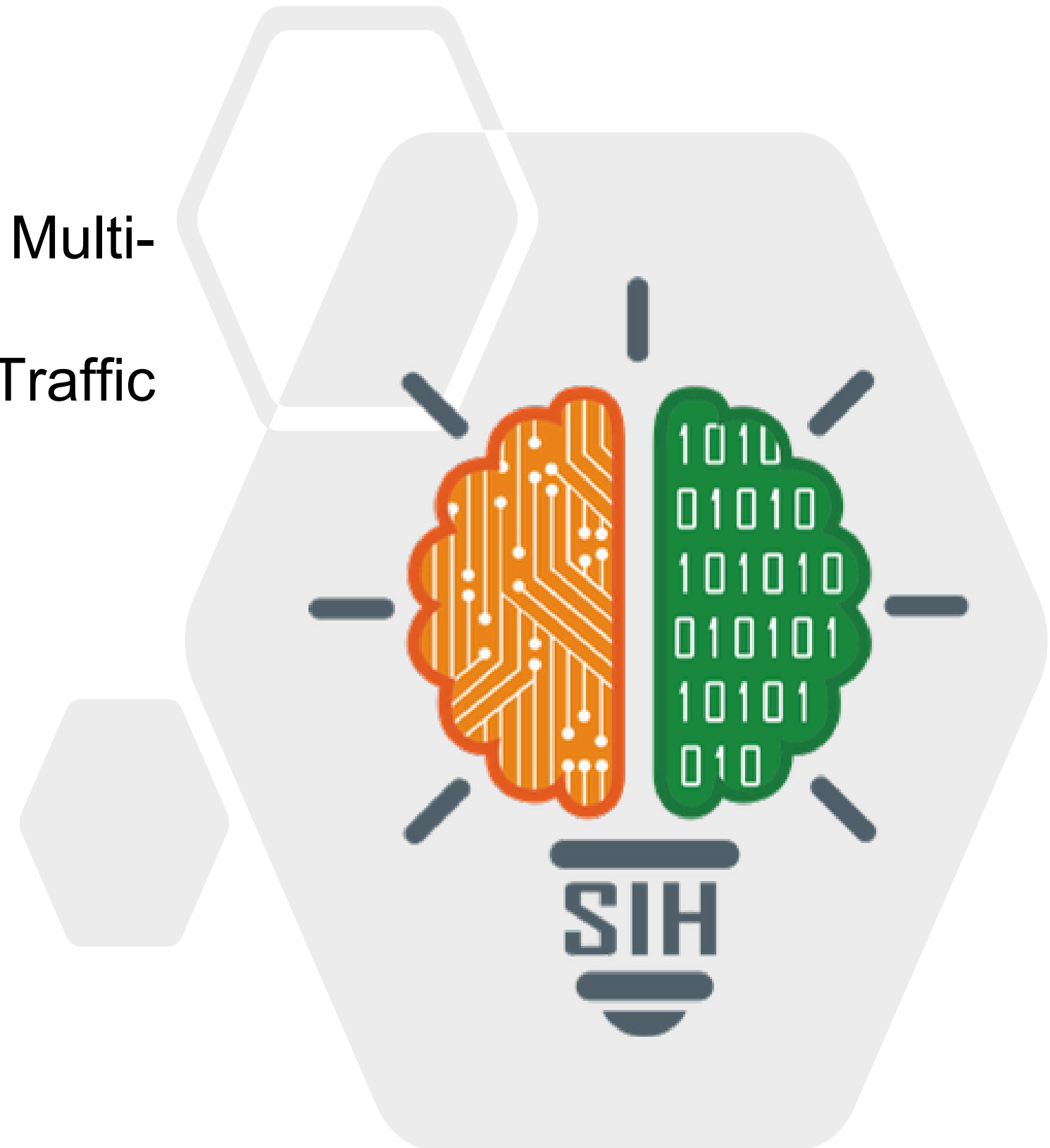


TITLE PAGE

- **Problem Statement ID** – SIH26127
- **Problem Statement Title**-City-Wide AI Engine for Multi-Camera ANPR Trajectory Tracking and Urban Traffic Analytics
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**-
- **Team Name** - Jobless Coders



THE PROBLEM

Isolated
Systems

Existing CCTV/ANPR systems operate in isolated silos, making cross-camera vehicle correlation difficult.

Unreliable
OCR

ANPR struggles with blur, low light, angled plates, poor weather and damaged plates, reducing OCR accuracy.

No City-Wide
Tracking

Lack of cross-camera integration prevents complete vehicle trajectories and city-wide traffic insights.

INNOVATION AND UNIQUENESS

Quality-Aware
OCR

- Smart quality filtering + enhancement for accurate plate recognition.

Cross-Camera
Tracking

- Temporal OCR voting reconstructs vehicle journeys across cameras.

GIS Traffic
Intelligence

- Temporal OCR voting reconstructs vehicle journeys across cameras.

OUR SOLUTION

1. Intelligent Detection

YOLO-based detection identifies vehicles and license plates from multi-camera CCTV feeds, while tracking assigns a unique Track ID.

2. Quality-Aware Processing

A Quality Gate evaluates blur, visibility and plate size, applying perspective correction & enhancement only when required.

3. Robust OCR

PaddleOCR + PARSeq generate plate predictions, with confidence-based consensus improving recognition in challenging conditions.

4. Temporal & Cross-Camera Tracking

Multi-frame voting + edit-distance alignment produces reliable plate numbers and links vehicles across geographically distributed cameras.

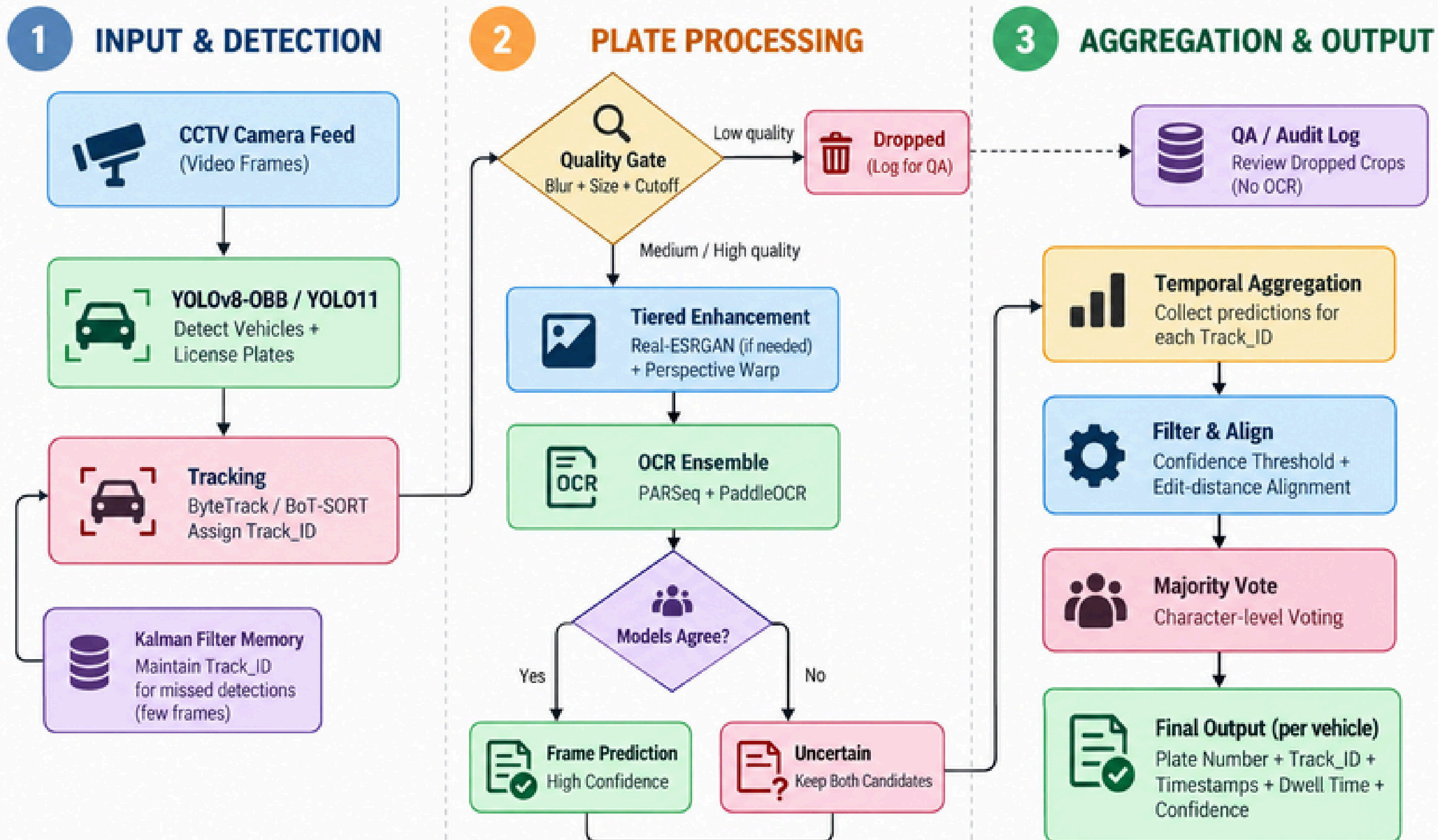
5. GIS Traffic Intelligence

A centralized GIS dashboard visualizes trajectories, traffic density, route patterns, congestion hotspots & heatmaps.

6. Real-Time Alerts

The platform automatically flags blacklisted vehicles and detects suspicious route anomalies for rapid intervention.

SYSTEM ARCHITECTURE



Tech Stack

Frontend

 **React**
 **Tailwind CSS**

 **Leaflet**
 **Machine Learning**

 **PaddleNET**
 **Open CV**

 **Py Torch**
 **UltraLytics**

Backend

 **SQL**
 **Python**

FEASIBILITY AND VIABILITY

Technical Feasibility

- **YOLOv8-OBB / YOLO11:** Unified vehicle + plate detection
- **ByteTrack / BoT-SORT:** Persistent tracking across frames
- **PARSeq + PaddleOCR:** Dual-model OCR for reliable recognition
- **Quality Gate + Voting:** Filters poor frames, improves accuracy



Financial Feasibility

- **Existing CCTV:** No major camera infrastructure required
- **Open-source Stack:** Low software licensing cost
- **Efficient Processing:** Quality gate reduces unnecessary compute
- **Scalable Deployment:** Easily supports additional cameras

Operational Feasibility

- **Automated Pipeline:** CCTV → Detection → OCR → Final plate
- **Track_ID Based:** Links plate predictions to vehicles
- **Fail-safe Tracking:** Kalman memory handles missed detections
- **QA Logging:** Uncertain & dropped cases retained for review



Smart Traffic Network
Real-time alerts and analytics



4

Phase 4

Phase 3



3

Multi-Camera
Automation +

Real-World Validation
Accuracy testing



2

Phase 2

Phase 1

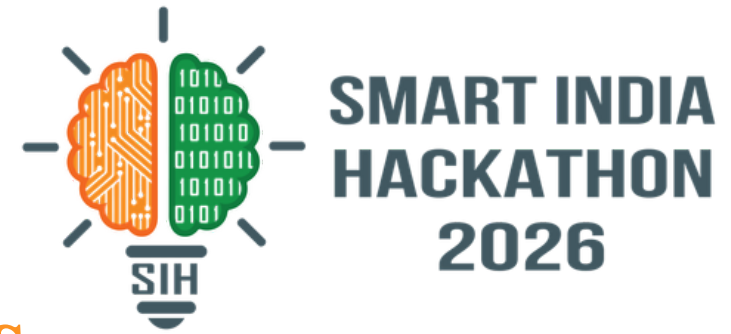


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End-to-End
Detection + OCR

Jobless
Coders

IMPACT AND BENEFITS



City Authorities

Improved ability to track high-interest or suspicious vehicles seamlessly across multiple zones.



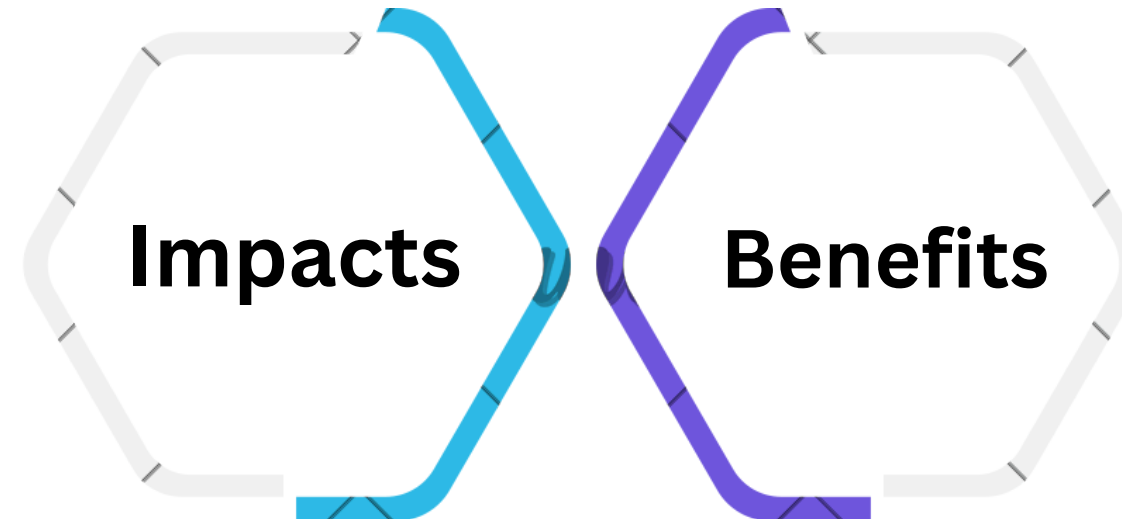
Social Benefits

- Increased public security by tracking blacklisted or suspicious vehicles.
- Improved emergency response (e.g., ambulances routed efficiently using traffic heatmaps).



Urban Planners

Access to macro-level traffic flow analytics for better infrastructure planning.



Economic Benefits

- Reduced fuel wastage and commuting time due to congestion management.
- Lower infrastructure costs by optimizing existing camera networks instead of deploying new ones.



Citizens

Enhanced road safety and quicker resolution of traffic bottlenecks.



Environmental Benefits

- Decreased carbon emissions from idling vehicles in traffic jams.
- Better urban air quality due to reduced congestion.

- Links of the reference and research work
 - CVPR – City-Scale Multi-Camera Vehicle Tracking
<https://openaccess.thecvf.com/>
 - Traffic-Aware Multi-Camera Vehicle Tracking
<https://arxiv.org>
 - Vehicle Re-Identification for Traffic Surveillance
<https://openaccess.thecvf.com>
 - Ultralytics YOLO Documentation
<https://docs.ultralytics.com>
 - PaddleOCR Documentation
<https://www.paddleocr.ai>
 - MapLibre GL JS Documentation
<https://maplibre.org>
 - OpenStreetMap – Routing & Road Networks
<https://www.openstreetmap.org>